

W4 – 4.5 Prove Trig Identities

MHF4U

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Prove each identity using the space on the following pages.

a) $\sin(x + y) = \sin x \cos y + \cos x \sin y$

c) $\sin(2x) = 2 \sin x \cos x$

e) $\cot \theta - \tan \theta = 2 \cot(2\theta)$

g) $\sin x \sec x = \tan x$

i) $\frac{\sec \theta - 1}{1 - \cos \theta} = \sec \theta$

k) $\frac{1 - \sin^2 x \cos^2 x}{\cos^4 x} = \tan^4 x + \tan^2 x + 1$

m) $\cot \theta - \tan \theta = 2 \cot(2\theta)$

o) $\frac{2 \tan x}{1 + \tan^2 x} = \sin(2x)$

q) $\cos^4 x - \sin^4 x = \cos(2x)$

s) $\cos(2x) = 2 \cos^2 x - 1$

u) $\frac{\cos(2x) + 1}{\sin(2x)} = \cot x$

b) $\tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$

d) $\cos(2x) = \cos^2 x - \sin^2 x$

f) $\frac{\sin(2\theta)}{1 - \cos(2\theta)} = \cot \theta$

h) $\frac{1 - \sin x}{\cos x} = \frac{\cos x}{1 + \sin x}$

j) $\frac{\sin x - \cos x}{\cos x} + \frac{\sin x + \cos x}{\sin x} = \sec x \csc x$

l) $\frac{\cos(2x) + 1}{\sin(2x)} = \cot x$

n) $(\sin x + \cos x)^2 = 1 + \sin(2x)$

p) $\sin\left(\frac{\pi}{4} + x\right) + \sin\left(\frac{\pi}{4} - x\right) = \sqrt{2} \cos x$

r) $\csc(2x) + \cot(2x) = \cot x$

t) $\sin\left(\frac{3\pi}{2} - x\right) = -\cos x$

v) $\cot x + \tan x = 2 \csc(2x)$

a) LS
$$\begin{aligned} &= \sin(x+y) \\ &= \sin[x - (-y)] \\ &= \sin x \cos(-y) - \cos x \sin(-y) \\ &= \sin x \cos y - \cos x (-\sin y) \\ &= \sin x \cos y + \cos x \sin y \end{aligned}$$

RS
$$= \sin x \cos y + \cos x \sin y$$

b) LS
$$\begin{aligned} &= \tan(x-y) \\ &= \tan[x + (-y)] \\ &= \frac{\tan x + \tan(-y)}{1 - \tan x \tan(-y)} \\ &= \frac{\tan x + (-\tan y)}{1 - \tan x (-\tan y)} \\ &= \frac{\tan x - \tan y}{1 + \tan x \tan y} \end{aligned}$$

RS
$$= \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

$$\begin{aligned}
 \text{c) } \quad \underline{Ls} &= \sin(2x) \\
 &= \sin(x+x) \\
 &= \sin x \cos x + \cos x \sin x \\
 &= 2 \sin x \cos x \\
 \underline{Rs} &= 2 \sin x \cos x
 \end{aligned}$$

$$\begin{aligned}
 \text{d) } \quad \underline{Ls} &= \cos(2x) \\
 &= \cos(x+x) \\
 &= \cos x \cos x - \sin x \sin x \\
 &= \cos^2 x - \sin^2 x \\
 \underline{Rs} &= \cos^2 x - \sin^2 x
 \end{aligned}$$

$$\begin{aligned}
 \text{e) } \quad \underline{Ls} &= \cot \theta - \tan \theta \\
 &= \frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta} \\
 &= \frac{\cos^2 \theta - \sin^2 \theta}{\cos \theta \sin \theta} \\
 \underline{Rs} &= 2 \cot(2\theta) \\
 &= \frac{2 \cos(2\theta)}{\sin(2\theta)} \\
 &= \frac{2 \cos(2\theta)}{2 \sin \theta \cos \theta} \\
 &= \frac{\cos^2 \theta - \sin^2 \theta}{\sin \theta \cos \theta}
 \end{aligned}$$

$$\begin{aligned}
 \text{f) } \quad \underline{Ls} &= \frac{\sin(2\theta)}{1 - \cos(2\theta)} \\
 &= \frac{2 \sin \theta \cos \theta}{1 - (1 - 2 \sin^2 \theta)} \\
 &= \frac{2 \sin \theta \cos \theta}{1 - 1 + 2 \sin^2 \theta} \\
 &= \frac{\cancel{2} \sin \theta \cos \theta}{\cancel{2} \sin^2 \theta} \\
 &= \frac{\cos \theta}{\sin \theta} \\
 \underline{Rs} &= \cot \theta \\
 &= \frac{\cos \theta}{\sin \theta}
 \end{aligned}$$

$$\begin{aligned}
 \text{g) } \quad \underline{Ls} &= \sin x \sec x \\
 &= \sin x \left(\frac{1}{\cos x} \right) \\
 &= \frac{\sin x}{\cos x} \\
 \underline{Rs} &= \tan x \\
 &= \frac{\sin x}{\cos x}
 \end{aligned}$$

$$\begin{aligned}
 \text{h) } \quad \underline{Ls} &= \frac{1 - \sin x}{\cos x} \\
 \underline{Rs} &= \frac{\cos x}{1 + \sin x} \times \frac{1 - \sin x}{1 - \sin x} \\
 &= \frac{\cos x (1 - \sin x)}{1 - \sin^2 x} \\
 &= \frac{\cancel{\cos x} (1 - \sin x)}{\cos^2 x} \\
 &= \frac{1 - \sin x}{\cos x}
 \end{aligned}$$

$$\begin{aligned}
 i) \quad & \underline{LS} \\
 & = \frac{\sec \theta - 1}{1 - \cos \theta} \\
 & = \frac{\frac{1}{\cos \theta} - \frac{\cos \theta}{\cos \theta}}{1 - \cos \theta} \\
 & = \left(\frac{1 - \cos \theta}{\cos \theta} \right) \\
 & = \left(\frac{1 - \cos \theta}{1} \right) \\
 & = \left(\frac{1 - \cancel{\cos \theta}}{\cos \theta} \right) \left(\frac{1}{\cancel{1 - \cos \theta}} \right) \\
 & = \frac{1}{\cos \theta}
 \end{aligned}$$

$$\begin{aligned}
 & \underline{RS} \\
 & = \sec \theta \\
 & = \frac{1}{\cos \theta}
 \end{aligned}$$

$$\begin{aligned}
 j) \quad & \underline{LS} \\
 & = \frac{\sin x}{\sin x} \frac{\sin x - \cos x}{\cos x} + \frac{\sin x + \cos x}{\sin x} \times \frac{\cos x}{\cos x} \\
 & = \frac{\sin x (\sin x - \cos x) + \cos x (\sin x + \cos x)}{\sin x \cos x} \\
 & = \frac{\sin^2 x - \cancel{\sin x \cos x} + \cancel{\cos x \sin x} + \cos^2 x}{\sin x \cos x} \\
 & = \frac{\sin^2 x + \cos^2 x}{\sin x \cos x} \\
 & = \frac{1}{\sin x \cos x}
 \end{aligned}$$

$$\begin{aligned}
 & \underline{RS} \\
 & = \sec x \csc x \\
 & = \left(\frac{1}{\cos x} \right) \left(\frac{1}{\sin x} \right) \\
 & = \frac{1}{\sin x \cos x}
 \end{aligned}$$

$$k) \quad \underline{LS} = \frac{1 - \sin^2 x \cos^2 x}{\cos^4 x}$$

$$\begin{aligned}
 & \underline{RS} \\
 & = \tan^4 x + \tan^2 x + 1 \\
 & = \frac{\sin^4 x}{\cos^4 x} + \frac{\sin^2 x}{\cos^2 x} \times \frac{\cos^2 x}{\cos^2 x} + \frac{1}{1} \times \frac{\cos^4 x}{\cos^4 x} \\
 & = \frac{\sin^4 x + \sin^2 x \cos^2 x + \cos^4 x}{\cos^4 x} \\
 & = \frac{(\sin^2 x)(\sin^2 x) + \sin^2 x \cos^2 x + (\cos^2 x)(\cos^2 x)}{\cos^4 x} \\
 & = \frac{\sin^2 x (1 - \cos^2 x) + \sin^2 x \cos^2 x + \cos^2 x (1 - \sin^2 x)}{\cos^4 x} \\
 & = \frac{\sin^2 x - \cancel{\sin^2 x \cos^2 x} + \cancel{\sin^2 x \cos^2 x} + \cos^2 x - \cancel{\sin^2 x \cos^2 x}}{\cos^4 x} \\
 & = \frac{\sin^2 x + \cos^2 x - \cancel{\sin^2 x \cos^2 x}}{\cos^4 x} \\
 & = \frac{1 - \sin^2 x \cos^2 x}{\cos^4 x}
 \end{aligned}$$

$$\begin{aligned}
 \text{l) } \quad & \underline{\text{LS}} \\
 &= \frac{\cos(2x) + 1}{\sin(2x)} \\
 &= \frac{2\cos^2 x - 1 + 1}{2\sin x \cos x} \\
 &= \frac{2\cos^2 x}{2\sin x \cos x} \\
 &= \frac{\cos x}{\sin x}
 \end{aligned}$$

$$\begin{aligned}
 \underline{\text{RS}} \\
 &= \cot x \\
 &= \frac{\cos x}{\sin x}
 \end{aligned}$$

$$\begin{aligned}
 \text{m) } \quad & \underline{\text{LS}} \\
 &= \cot \theta - \tan \theta \\
 &= \frac{\cos \theta}{\sin \theta} \times \frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta} \times \frac{\sin \theta}{\sin \theta} \\
 &= \frac{\cos^2 \theta - \sin^2 \theta}{\cos \theta \sin \theta}
 \end{aligned}$$

$$\begin{aligned}
 \underline{\text{RS}} \\
 &= 2\cot(2\theta) \\
 &= \frac{2\cos(2\theta)}{\sin(2\theta)} \\
 &= \frac{2\cos(2\theta)}{2\sin \theta \cos \theta} \\
 &= \frac{\cos^2 \theta - \sin^2 \theta}{\sin \theta \cos \theta}
 \end{aligned}$$

$$\begin{aligned}
 \text{n) } \quad & \underline{\text{LS}} \\
 &= (\sin x + \cos x)^2 \\
 &= (\sin x + \cos x)(\sin x + \cos x) \\
 &= \sin^2 x + \sin x \cos x + \cos x \sin x + \cos^2 x \\
 &= \sin^2 x + \cos^2 x + \sin x \cos x + \sin x \cos x \\
 &= 1 + 2\sin x \cos x
 \end{aligned}$$

$$\begin{aligned}
 \underline{\text{RS}} \\
 &= 1 + \sin(2x) \\
 &= 1 + 2\sin x \cos x
 \end{aligned}$$

$$\begin{aligned}
 \text{o) } \quad & \underline{\text{LS}} \\
 &= \frac{2\tan x}{1 + \tan^2 x} \\
 &= \frac{2\left(\frac{\sin x}{\cos x}\right)}{1 + \frac{\sin^2 x}{\cos^2 x}} \\
 &= \left(\frac{2\sin x}{\cos x}\right) \frac{\cos^2 x}{\cos^2 x + \frac{\sin^2 x}{\cos^2 x}}
 \end{aligned}$$

$$\begin{aligned}
 \underline{\text{RS}} \\
 &= \sin(2x) \\
 &= 2\sin x \cos x
 \end{aligned}$$

$$\begin{aligned}
 &= \left(\frac{2\sin x}{\cos x}\right) \frac{1}{\left(\frac{1}{\cos^2 x}\right)} \\
 &= \left(\frac{2\sin x}{\cancel{\cos x}}\right) \left(\frac{\cos^{\cancel{1}} x}{1}\right) \\
 &= 2\sin x \cos x
 \end{aligned}$$

$$\begin{aligned}
 \text{p) } \quad & \underline{\text{LS}} \\
 &= \sin\left(\frac{\pi}{4} + x\right) + \sin\left(\frac{\pi}{4} - x\right) \\
 &= \sin \frac{\pi}{4} \cos x + \cancel{\cos \frac{\pi}{4} \sin x} + \sin \frac{\pi}{4} \cos x - \cancel{\cos \frac{\pi}{4} \sin x} \\
 &= 2\sin \frac{\pi}{4} \cos x \\
 &= 2\left(\frac{\sqrt{2}}{2}\right) \cos x \\
 &= \sqrt{2} \cos x
 \end{aligned}$$

$$\begin{aligned}
 \underline{\text{RS}} \\
 &= \sqrt{2} \cos x
 \end{aligned}$$

$$\begin{aligned}
 \text{q) } \quad & \underline{\text{LS}} \\
 &= \cos^4 x - \sin^4 x \\
 &= (\cos^2 x)^2 - (\sin^2 x)^2 \\
 &= (\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x) \\
 &= (\cos^2 x - \sin^2 x)(1) \\
 &= \cos^2 x - \sin^2 x
 \end{aligned}$$

$$\begin{aligned}
 \underline{\text{RS}} \\
 &= \cos(2x) \\
 &= \cos^2 x - \sin^2 x
 \end{aligned}$$

$$\begin{aligned}
 r) \quad & \underline{LS} \\
 & = \csc(2x) + \cot(2x) \\
 & = \frac{1}{\sin(2x)} + \frac{\cos(2x)}{\sin(2x)} \\
 & = \frac{1 + \cos(2x)}{\sin(2x)} \\
 & = \frac{1 + 2\cos^2 x - 1}{2\sin x \cos x} \\
 & = \frac{\cancel{2}\cos^2 x}{\cancel{2}\sin x \cancel{\cos x}} \\
 & = \frac{\cos x}{\sin x}
 \end{aligned}$$

$$\begin{aligned}
 & \underline{RS} \\
 & = \cot x \\
 & = \frac{\cos x}{\sin x}
 \end{aligned}$$

$$\begin{aligned}
 s) \quad & \underline{LS} \\
 & = \cos(2x) \\
 & = \cos(x+x) \\
 & = \cos x \cos x - \sin x \sin x \\
 & = \cos^2 x - \sin^2 x \\
 & = \cos^2 x - (1 - \cos^2 x) \\
 & = \cos^2 x - 1 + \cos^2 x \\
 & = 2\cos^2 x - 1 \\
 & \underline{RS} \\
 & = 2\cos^2 x - 1
 \end{aligned}$$

$$\begin{aligned}
 t) \quad & \underline{LS} \\
 & = \sin\left(\frac{3\pi}{2} - x\right) \\
 & = \sin \frac{3\pi}{2} \cos x - \cos \frac{3\pi}{2} \sin x \\
 & = -1 \cos x - 0 \sin x \\
 & = -\cos x \\
 & \underline{RS} \\
 & = -\cos x
 \end{aligned}$$

$$\begin{aligned}
 u) \quad & \underline{LS} \\
 & = \frac{\cos(2x) + 1}{\sin(2x)} \\
 & = \frac{2\cos^2 x - \cancel{1} + \cancel{1}}{2\sin x \cos x} \\
 & = \frac{\cancel{2}\cos^2 x}{\cancel{2}\sin x \cancel{\cos x}} \\
 & = \frac{\cos x}{\sin x} \\
 & \underline{RS} \\
 & = \cot x \\
 & = \frac{\cos x}{\sin x}
 \end{aligned}$$

$$\begin{aligned}
 v) \quad & \underline{LS} \\
 & = \cot x + \tan x \\
 & = \frac{\cancel{\cos x}}{\cancel{\cos x}} \times \frac{\cos x}{\sin x} + \frac{\sin x}{\cos x} \times \frac{\cancel{\sin x}}{\cancel{\sin x}} \\
 & = \frac{\cos^2 x + \sin^2 x}{\cos x \sin x} \\
 & = \frac{1}{\sin x \cos x} \\
 & \underline{RS} \\
 & = 2 \csc(2x) \\
 & = \frac{2}{\sin(2x)} \\
 & = \frac{\cancel{2}}{\cancel{2}\sin x \cos x} \\
 & = \frac{1}{\sin x \cos x}
 \end{aligned}$$